

Arnav Sharma

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EDUCATION

University of California, Davis

Expected Jun 2027

B.S. in Computer Science · Minor in Business Studies

Dean's List: Spring 2024

Relevant Coursework: Machine Learning, Artificial Intelligence, Computer Vision, Operating Systems, Computer Architecture

EXPERIENCE

Fonabit AI

Jun 2026 – Sep 2026

Software Engineer Intern

- Developed custom Model Context Protocol (MCP) servers and APIs that let LLM agents query and act on enterprise CRM data.
- Architected multi-agent orchestration over Agent-to-Agent (A2A) protocols, automating sales and lead-management workflows end to end.
- Hardened the agent layer with prompt-injection defense, role-based access control, and human-in-the-loop approval gates on high-stakes actions.
- Cut token overhead and response latency by restructuring prompt chains and context-window retrieval.

Pixabits Technologies Pvt. Ltd.

Jun 2025 – Aug 2025

Software Developer Intern

- Engineered backend services and REST APIs in Python and Java, cutting response times and lifting throughput over 50%.
- Profiled and eliminated performance bottlenecks alongside cross-functional teams, shipping fixes across multiple releases.
- Converted Figma designs into responsive UI components and refined them against user analytics.

PROJECTS

Credere AI — Agentic Credit-Note & Reconciliation Platform

2026 · Site · Demo

- Designed and deployed an agentic accounting platform that ingests a company's bank feed, classifies every payment, issues the credit owed, and writes it back to their ledger; 19 containerized services running live against ~870 passing tests.
- Constrained the LLM to an advisory role: every proposal clears schema, invoice, and policy validation before execution, and an AST-level test proves no agent module can import an issuance capability.
- Enforced multi-tenant isolation in PostgreSQL with FORCE row-level security and a NOBYPASSRLS role, backed by tests asserting unscoped queries fail closed instead of leaking across tenants.
- Eliminated double-posting under concurrent workers and mid-flight crashes using idempotency keys, a transactional outbox, and claim-with-lease drainers.
- Integrated Zoho Books (OAuth 2.0), Plaid, and ISO 20022 camt.053 behind a single adapter contract, and shipped a React/TypeScript operator console.

CardioSense

2026 · HackDavis · GitHub

- Built a sub-\$30 wearable ECG that detects atrial fibrillation entirely on-device, returning a rhythm verdict in under 10 seconds with no cloud dependency.
- Trained a 43K-parameter 1D CNN (PyTorch) on MIT-BIH clinical data to ~91% accuracy, rising to 95% behind a strict "Uncertain" abstention gate, at ~10 ms inference.
- Implemented the full signal chain (filtering, Pan-Tompkins R-peak detection, HRV feature extraction), feeding real-time SwiftUI and Streamlit clients over BLE.

InfraCopilot AI

2026 · UC Davis · GitHub

- Developed a predictive-maintenance platform for EV charging networks around a cost-aware model (scikit-learn, SMOTE) reaching ~90% failure recall and \$300K+ in simulated savings.
- Engineered the FastAPI backend and Next.js dashboard behind it, surfacing real-time risk scores and root-cause analysis.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, Haskell, Erlang, Prolog, Assembly

Machine Learning & AI: PyTorch, scikit-learn, CNNs, Transformers, LLM Agents (Claude/Gemini API, LangGraph, MCP)

Paradigms & Systems: Functional Programming, Logic Programming, Concurrency, Operating Systems, Memory Management

Backend & Tools: PostgreSQL (RLS), FastAPI, Docker, REST APIs, React/TypeScript, Git, Linux, pytest